

Project Report: Information Retrieval Engine

Introduction

This project entails the creation of an Information Retrieval (IR) engine that generates forward and inverted indices. Documents are scanned by the IR engine in order to support indexing of documents for search terms defined by the user. The components of this project include the TextProcessor, DocIndex, InvertedIndex, QueryProcessor, and the main IR engine.

Design and Implementation

Text Processor

The TextProcessor class is responsible for processing the actual content of the documents. This includes removing stop words, as well as stemming, using the Porter stemmer. The tokens processed will be forwarded to the indexing process.

DocIndex

The DocIndex class maintains a forward index, mapping the docID to word frequency. This index will help understand the distribution frequency of terms in the document.

InvertedIndex

The InvertedIndex class maintains an inverted index that maps words to docID and frequency. This index allows more efficient retrieval of documents that contain terms searched by a user.

QueryProcessor

The QueryProcessor class uses the Inverted Index to retrieve search terms and return documents.

A method to process the query is available that returns the query terms as well as their respective frequency.

Main

The Main class controls and coordinates the entire process.

The Main class loads stop words, initializes the Porter stemmer and text processor, reads input documents.

It is also responsible for returning user queries and displaying a return to the search.

Compile Java Files:

```
javac -d bin src/com/ir/*.java
```

Run the Main Class:

```
java -cp bin com.ir.Main
```